AtiNAMatiC-Air — User Manual

Version 0.2.0 · Atij 666 Studio · Windows (Standalone & VST3)

What's new in v0.2.0

- **Real depth on mono sources.** A guitar coming out of a mono amp sim has no side signal, so the old M/S widening produced no stereo image at all. Air now builds its depth from a short, mono-safe early-reflection pattern written only into the side channel (Studio: 2–12 ms, Room: 5–29 ms, lows below 150 Hz kept centred). Summing to mono gives exactly the dry-plus-EQ signal back. It is still not a reverb — there is no tail.
- The integer-step Haas delay (which zippered when the knob moved) is gone; the depth meter now shows the side-to-mid energy you actually hear.
- **New interface** — machined knob with a printed 0–10 scale, hardware-style mode switches that follow automation and preset recall, consistent value text ("72 %", "+1.5 dB"), resizable window (75–150 %, remembered per session). Double-click any knob to reset it; hold **Shift** while dragging for fine adjustment.

1. What it is

AtiNAMatiC-Air is a post-cab "amp-in-the-room" enhancer designed to sit **after** your amp/cab simulator (NAM, Helix Native, ML Sound Lab, IRs, etc.) in the signal chain. It restores three things that get lost when a cab is captured through a single microphone:

1. **Air** — high-shelf lift (10 kHz) plus a 5 kHz presence peak that brings the cab out of the speaker and into the room.
2. **Body** — a gentle 80 Hz peak (Room mode only) that adds the low-mid bloom you hear when standing in front of a real 4×12.
3. **Width & depth** — M/S widening plus a short, mono-safe early-reflection pattern in the side channel (more in Room mode), so even a mono amp sim gets a three-dimensional image.

It is **not** a reverb. It does not add tail. It re-creates the perceptual cues that make a recorded cab feel like a real amp in a real room.

2. Signal chain placement

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Guitar → Amp Sim (NAM / Helix / etc.) → Cab / IR → ★ AtiNAMatiC-Air ★ → Reverb / Delay → Master
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Always place AtiNAMatiC-Air after the cab/IR, never before. It is voiced for cab-coloured material, not raw amp distortion.

3. The Controls

AIR (large knob, centre)

The master macro. From 0 to 100% it simultaneously increases: - High-shelf gain at 10 kHz (0 → +5 dB) - Presence peak at 5 kHz (0 → +2 dB) - Body peak at 80 Hz (Room mode only, 0 → +2.5 dB) - Stereo widening (Studio: +0–40% sides · Room: +0–70% sides) - Early-reflection depth in the side channel (Studio: 2–12 ms, Room: 5–29 ms), mono-safe

Most users live between **30% and 65%**. Above 70% the widening becomes deliberately exaggerated for "WIDE" mix moments.

MODE (Studio / Room)

- **STUDIO** — Tighter, narrower, no body peak, a few very short reflections. Use for dry mixes, tight metal rhythms, double-tracked guitars where mono compatibility matters.
- **ROOM** — Adds the 80 Hz body peak and a longer early-reflection pattern that recreates the ear-distance cues of standing in a real room. Use for leads, ambient parts, and any track that needs to feel "live".

MIX (Wet/Dry)

0% = bypass, 100% = full wet. Default 100%. Lower it to blend in just a touch of air without changing the EQ balance of your dry cab.

OUTPUT (–24 to +12 dB)

Final makeup gain after all processing. Use to compensate for level changes the EQ + widening produces.

Activity Pill (bottom right)

Real-time indicator showing how much "air" is currently being applied:

Label	Meaning
DRY	Knob at 0, or no signal coming in
LIFT	Subtle enhancement (10–40%)
OPEN	Standard amp-in-room (40–70%)
WIDE	Wide-stereo / mix-bus mode (70–100%)

The LED brightens as you play and dims as the signal decays — it tracks both **the AIR knob position and the input level** so it idles dim when no notes are being played.

Activity Meter (INPUT / OUTPUT)

Two horizontal level bars showing peak input (post-cab) and post-processing output, with a dB scale below from –60 dB to 0 dB.

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4. Quick-start presets (manual recipes)

Tight rhythm (mono-compatible)

- MODE: Studio · AIR: 35% · MIX: 100% · OUT: 0 dB

Standard rhythm (room feel)

- MODE: Room · AIR: 45% · MIX: 100% · OUT: −1 dB

Solo / lead

- MODE: Room · AIR: 65% · MIX: 100% · OUT: 0 dB

Ambient / clean

- MODE: Room · AIR: 80% · MIX: 100% · OUT: −2 dB

Mix-bus glue (parallel)

- MODE: Studio · AIR: 25% · MIX: 35% · OUT: 0 dB

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5. Plug-in formats & install paths

Format	Default install location
Standalone	`C:\Program Files\Atij 666 Studio\AtiNAMatiC-Air\AtiNAMatiC-Air.exe`
VST3	`C:\Program Files\Common Files\VST3\AtiNAMatiC-Air.vst3`

After install, **rescan plug-ins in your DAW**.

- **Reaper:** Options → Preferences → VST → Re-scan
- **Cubase / Nuendo:** Studio → VST Plug-in Manager → Update
- **Studio One:** Options → Locations → VST Plug-ins → Reset Blacklist + Rescan
- **Ableton Live:** Preferences → Plug-Ins → Use VST3 Plug-in Custom Folder → Rescan

In your DAW the plug-in appears under manufacturer **Atij666Studio**, category **Fx / EQ / Guitar**.

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6. CPU / latency

- Processing is **zero-latency** (no oversampling, no FFT, no look-ahead).
- IIR filtering only: < 1 % CPU on a modern machine, even at 96 kHz.
- Safe to instantiate dozens of times in one session.

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7. Troubleshooting

“The Activity Pill never lights up beyond DRY”
Make sure something is actually feeding the plug-in (NAM upstream, track armed/playing) and the AIR knob is above 0. The pill measures air × input level — both must be non-zero.

“It sounds harsh”
Drop AIR. Above 70% with a bright cab IR the 10 kHz shelf can stack up. Try 35–50% first.

“It feels narrower than expected”
Check that your DAW track (or the insert) is stereo. Air creates its depth from a mono source, but it needs a stereo output to put it in; on a mono track only the EQ is heard.

“Plug-in is missing after install”
Force a VST3 rescan in your DAW (see section 5). Some hosts cache old scan results.

“Mode switch produces a click”
Toggling MODE updates the body filter coefficients. The change is per-block so it should be inaudible during continuous playback; if you automate it, slope the automation across a few ms to be safe.

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8. Credits

- Design & development: **Atij** (Atij 666 Studio)
- Built with **JUCE 8** (juce.com)
- Inspired by the room sound of every great cab anyone ever stood in front of

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9. License & support

Personal & commercial use permitted on machines you own.
Bug reports / feature requests: see About dialog (click the title in the plug-in window).

AtiNAMatiC-Air is part of the AtiNAMatiC Series — a family of focused, hardware-flavoured guitar tools for the modern post-amp signal chain.